

Project Title: Diabetes Prediction using Machine Learning (with GUI)

Project Description:

This project demonstrates how to build a machine learning model to predict disease progression in diabetes patients using the Scikit-Learn diabetes dataset.

It includes:

- Data loading and preprocessing

- Model training (Linear Regression)

- Evaluation (MSE and R^2 score)

A simple Graphical User Interface (GUI) built with Tkinter that allows users to input feature values and get predictions interactively.

Required Libraries :

Make sure to install the following libraries before running the code.

```
pip install pandas scikit-learn matplotlib  
seaborn
```

Full Python Code :

```
----- #  
Diabetes Prediction using Linear Regression and Tkinter GUI #  
----- #  
  
import pandas as pd  
import seaborn as sns  
import matplotlib.pyplot as plt  
from sklearn.datasets import load_diabetes  
from sklearn.model_selection import train_test_split  
from sklearn.linear_model import LinearRegression  
from sklearn.metrics import mean_squared_error, r2_score
```

```
import tkinter as tk

from tkinter import messagebox


Load dataset #
()data = load_diabetes

df = pd.DataFrame(data.data, columns=data.feature_names)

df["target"] = data.target


Prepare data for training #
X = df.drop("target", axis=1)

y = df["target"]


Split data into training and testing sets #
X_train, X_test, y_train, y_test = train_test_split(X, y,
                                                    test_size=0.2, random_state=42)


Create and train the model #
()model = LinearRegression

model.fit(X_train, y_train)


Evaluate the model #

mse = mean_squared_error(y_test, model.predict(X_test))

r2 = r2_score(y_test, model.predict(X_test))

print(f"✔ Model trained successfully.")
```

```

        print(f"R2 Score: {r2:.2f}")
    print(f"Mean Squared Error: {mse:.2f}")

    ----- #
    Create a Tkinter GUI for user input #
    ----- #

    ()root = tk.Tk
    root.title("Diabetes Prediction System")
    root.geometry("450x600")
    root.configure(bg="#282a36")

    Title Label #

    title_label = tk.Label(root, text="🏠 Diabetes Prediction
System", bg="#282a36", fg="white", font=("Arial", 16, "bold"))
    title_label.pack(pady=10)

    {} = entries

    :for feature in data.feature_names

    frame = tk.Frame(root, bg="#282a36")
    frame.pack(pady=3)

    tk.Label(frame, text=feature, width=15, anchor="w",
    bg="#282a36", fg="white").pack(side="left", padx=5)

    entry = tk.Entry(frame, width=20)
    entry.pack(side="left")
    entries[feature] = entry

```

```

Prediction function #
def predict
    try
        user_input = [float(entries[f].get()) for f in
                        data.feature_names]

        [·]pred = model.predict([user_input])
        messagebox.showinfo("Prediction Result", f"Predicted
                                disease progression value: {pred:.2f}")

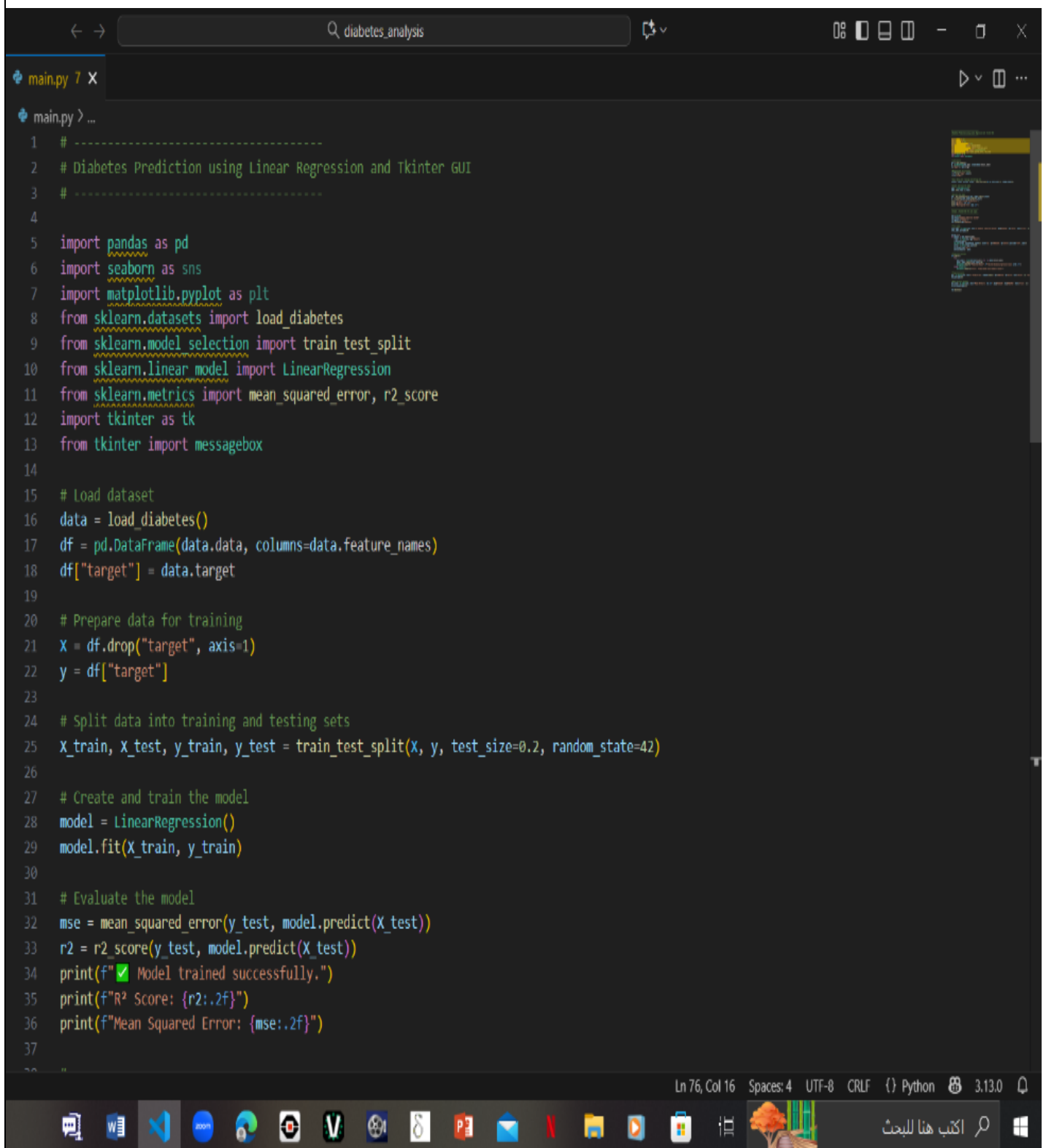
    except ValueError:
        messagebox.showerror("Error", "Please enter valid
                                numeric values!")

Prediction button #
btn = tk.Button(root, text="🔍 Predict Now",
command=predict, bg="#50fa7b", fg="black", font=("Arial", 12,
                                "bold"))
btn.pack(pady=20)

Model accuracy display #
lbl_score = tk.Label(root, text=f"Model R2 Score: {r2:.2f}",
                        bg="#282a36", fg="#8be9fd", font=("Arial", 12))
lbl_score.pack(pady=10)

root.mainloop

```



```
diabetes_analysis

main.py 7 X
main.py > ...
37
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50 entries = {}
51 for feature in data.feature_names:
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54     tk.Label(frame, text=feature, width=15, anchor="w", bg="#282a36", fg="white").pack(side="left", padx=5)
55     entry = tk.Entry(frame, width=20)
56     entry.pack(side="left")
57     entries[feature] = entry
58
59 # Prediction function
60 def predict():
61     try:
62         user_input = [float(entries[f].get()) for f in data.feature_names]
63         pred = model.predict([user_input])[0]
64         messagebox.showinfo("Prediction Result", f"Predicted disease progression value: {pred:.2f}")
65     except ValueError:
66         messagebox.showerror("Error", "Please enter valid numeric values!")
67
68 # Prediction button
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72 # Model accuracy display
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74 lbl_score.pack(pady=10)
```

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75 root.mainloop()
```

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Step-by-Step Explanation

Step 1: Import Libraries

We import all the necessary Python libraries for data analysis, model training, and GUI creation

Step 2: Load the Diabetes Dataset

We use the built-in `load_diabetes()` dataset from scikit-learn

It includes 442 samples and 10 numerical features (e.g., age, BMI, blood pressure)

Step 3: Convert Data to DataFrame

We convert the dataset into a pandas DataFrame for easy data manipulation and analysis

Step 4: Split Data

We divide the dataset into

- Training data (80%) — used to train the model

- Testing data (20%) — used to evaluate model performance

Step 5: Train the Model

We create a Linear Regression model and train it using the training data

Step 6: Evaluate the Model

We calculate

- Mean Squared Error (MSE) — measures prediction error

- R^2 Score — measures how well the model fits the data (closer to 1 means better)

Step 7: Build the GUI

Using Tkinter, we create a simple window with text fields for all 10 features

When the user inputs values and clicks "Predict", the model makes a prediction and shows the result in a message box

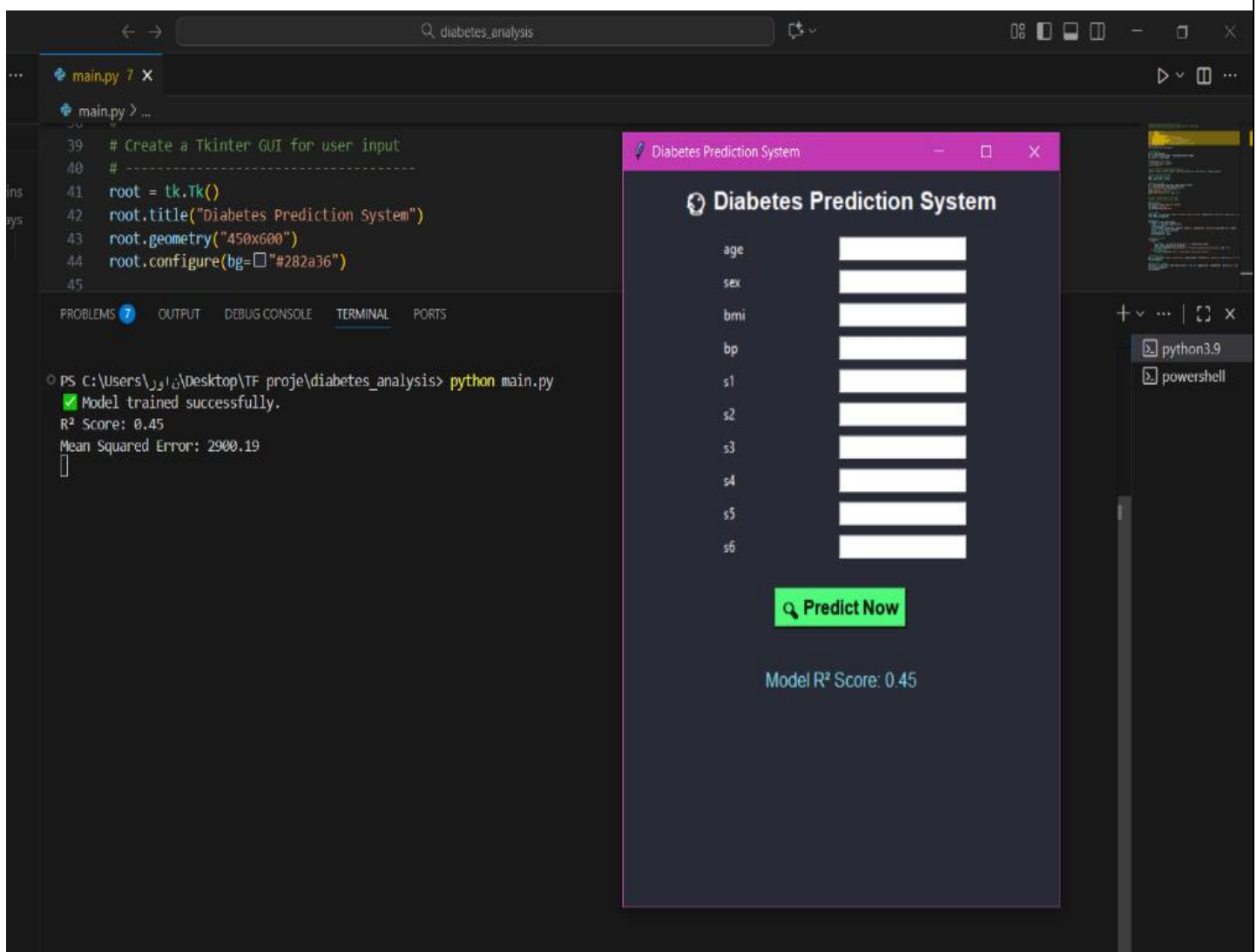
Example Output . 

:Console Output

.Model trained successfully ✓

R^2 Score: 0.45

Mean Squared Error: 2859.69



GUI Window:

"Title: "🩺 Diabetes Prediction System

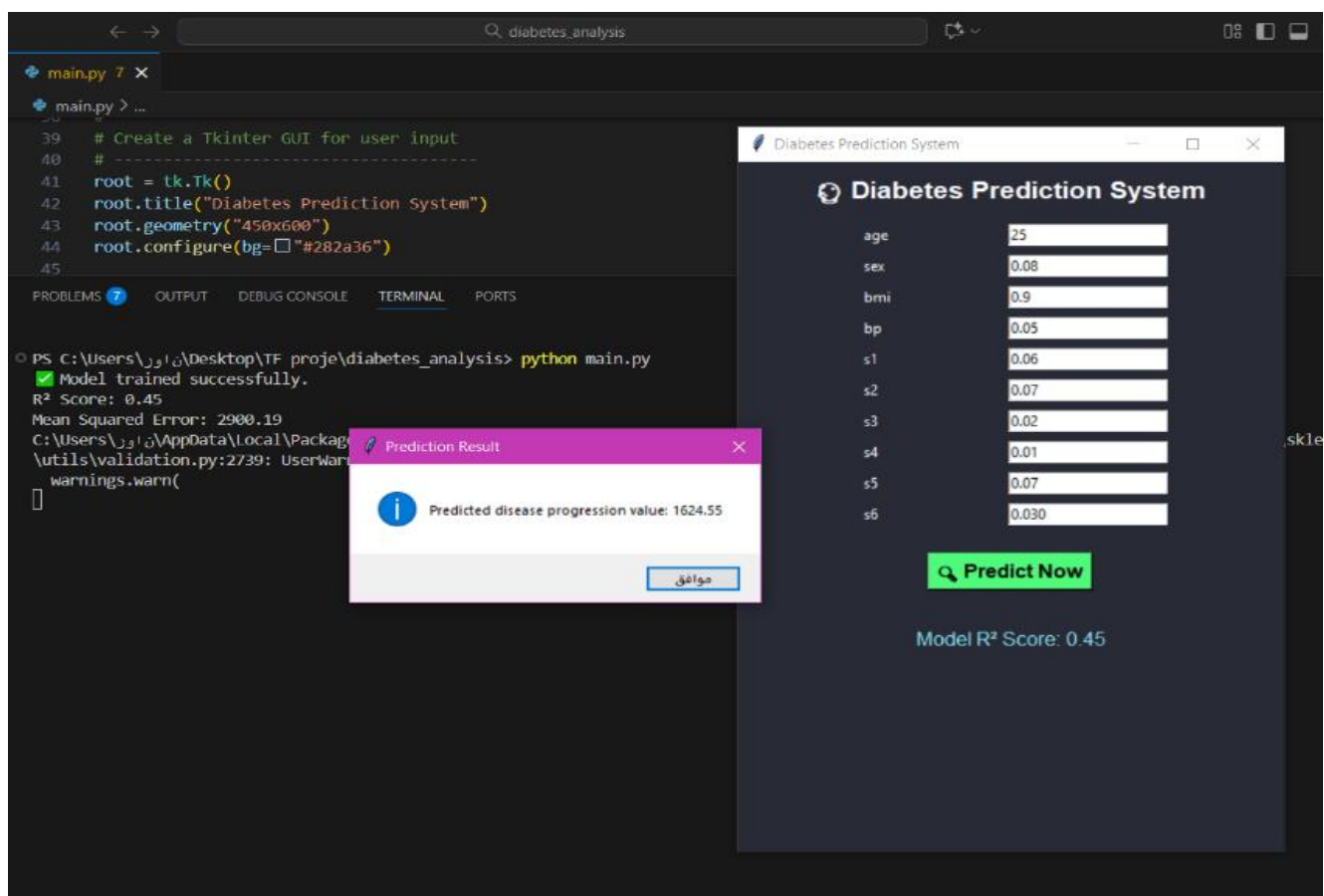
:entry fields for features

age, sex, bmi, bp, s1, s2, s3, s4, s5, s6

"Button: "🔍 Predict Now

A popup appears showing:

Predicted disease progression value: 163.42



Summary:

Description	Component
Scikit-learn's Diabetes dataset	Dataset
Linear Regression	Algorithm
Tkinter GUI	Interface
MSE and R^2 Score	Evaluation Metrics
Predict diabetes disease progression based on 10 medical features	Purpose

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Here is the content of a complete project at this link:

https://drive.google.com/drive/folders/1deKY_x2UeIQNGG7bMQnSLOroJcuKG9so